1. There are two nodes in a network that use stable storage and acknowledgment message passing for reliable communication. The stable storage of Node A contains the following record –
   Received message (In6); Transmitted message(Out3); Out:3 Ack:3 In:6.
   The stable storage of Node B contains the following record –
   Received message (In3); Transmitted message(Out6); Out:6 Ack:6 In:3.

   Now Node B sends a new message 7 to Node A. What will be in the stable storage of A and B if the message is received correctly, including a correctly received acknowledgement? [2 marks]

2. Assume a large distributed system has many servers at different locations. The network connection between the servers is not reliable. There are millions of active users who regularly use this system. Why eventual consistency is more suitable than strong consistency for this system? [4 marks]

3. Consider a database storing the records of one million families, where each family is a row in a table. A unique ID, the location of the family's current residence, and the total household income per year of each family are stored in the database. The total household income of any family can be between 0 to 2 million, and it is an integer number (i.e., a whole number without any decimal value). For each of the following queries on this database, which index is the most suitable choice for efficiency? If there is any search key that needs to be specified for that index, mention the search key as well. If there's no index that can improve a query's performance, write 'no index' for that query. You need to explain the rationale of each of your answers. [6 marks]

Answer for each of the following queries -

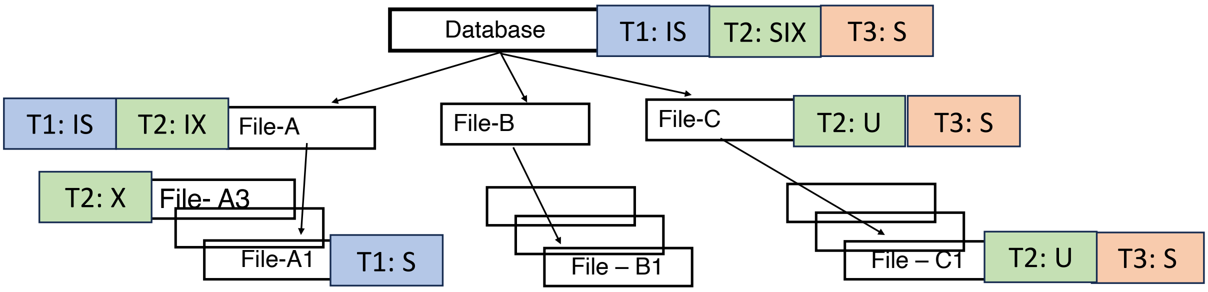
(i) find the IDs of the families where the total household income is less than 70,000;
(ii) find the household income of the families who live within 5km of Melbourne central station.
(iii) find the average household income of the families.

4. When a database needs to join two tables using a page-oriented nested loop join algorithm, both tables are found to be in the main memory. Will it change how the cost of a cost-based optimiser calculates the cost of the query plan? [2 marks]

5. Given the hierarchy of database objects and the corresponding granular locks in the following picture, which transactions can run concurrently from the beginning till end (that is, all operations and locks are compatible to run concurrently with another one) and which ones need to be delayed, answer for both scenarios –

   (i) Scenario 1: If the transactions arrive in the order T1-T2-T3. Explain your answer.
   (ii) Scenario 2: If the transactions arrive in the order T1-T3-T2. Explain your answer.

   Note that all transactions need to take the locks when they start to run. [6 marks]



A granular lock compatibility matrix is as follows –

Compatibility Mode of Granular Locks							
Current	None	IS	IX	S	SIX	U	X
Request	+/- (Next mode) + granted / - delayed						
IS	+(IS)	+(IS)	+(IX)	+(S)	+(SIX)	-(U)	-(X)
IX	+(IX)	+(IX)	+(IX)	-(S)	-(SIX)	-(U)	-(X)
S	+(S)	+(S)	-(IX)	+(S)	-(SIX)	-(U)	-(X)
SIX	+(SIX)	+(SIX)	-(IX)	-(S)	-(SIX)	-(U)	-(X)
U	+(U)	+(U)	-(IX)	+(U)	-(SIX)	-(U)	-(X)
X	+(X)	-(IS)	-(IX)	-(S)	-(SIX)	-(U)	-(X)